Arithmetic Function Analogs for K-Tuples

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Abstract

We encode the essential information of k-tuples in some extensions of common arithmetic functions. Analogs of some fundamental relationships between those common functions are shown to hold for our extensions. The basic analytic properties of some of these functions are derived and related to the first Hardy-Littlewood conjecture.

1 K-Tuples

Definition 1.1. *Define $\mathbb{K}$, the set of **k-tuples**, by*

$$\mathbb{K} := \mathscr{P}(\mathbb{Z}).$$

*We say a k-tuple is **singular** if it contains exactly one element.*

Some authors forbid infinite k-tuples. We denote the number of elements in a set S by $\#S$. The motivation for much of this paper is to extend a definition or result pertaining to integers to the k-tuples. That is, we view traditional number theoretic concepts involving an integer k as being about the singular k-tuple $\{0\}$, and attempt to extend that result in a natural way to nonsingular k-tuples. We also refer to such singular definitions and results collectively as "singular number theory".

Definition 1.2. *For $A, B \in \mathbb{K}$, we define*

$$A + B := \{a + b : a \in A, b \in B\}.$$

For $k \in \mathbb{Z}$ and $A \in \mathbb{K}$, define for convenience $k + A := \{k\} + A$. That is, $k + A$ is the translation of A by k.

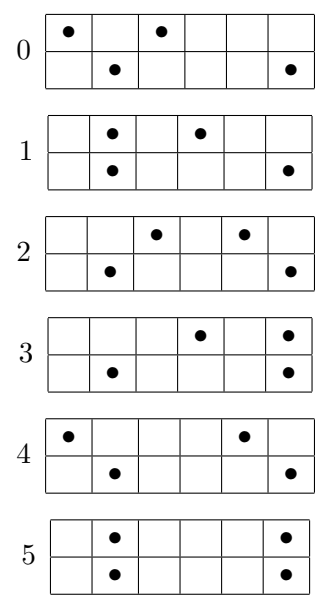
We are often concerned with congruence between elements of k-tuples, but the usual definition of congruence is insufficient, as it describes only exact incidence between numbers relative to a modulus. We are instead concerned with a variety of types of incidence between translated k-tuples and the integers relatively prime to some modulus. The following definition handles the "modulus" aspect of that need.

Definition 1.3. *Given a k-tuple A and a natural number n, define*

$$A \bmod n \text{ or } A_n := \{a \bmod n : a \in A\}$$

2 The Totient Function

We now extend Euler's totient function φ to k-tuples. Instead of counting which natural numbers up to n are relatively prime to n, we count the translates of a k-tuple A by $k \in \mathbb{Z}_n$ for which all elements are coprime to n. We may also consider counting many types of occurences, and will do so in a later section. We graph the counting process below for $A = \{0, 2\}$ and $n = 6$. The top grid represents a translate of A, and the bottom are those integers up to 6 coprime to 6. Recalling that those coprime integers are periodic with period 6, we simply wrap any elements of the translated A which fall off the end back to 0.



There is only a single occurrence, on table 5, of complete containment, so our totient should return 1 for this A and n.

Definition 2.1 (K-tuple totient)**.**
For a k-tuple A, define φ_A by:

$$\varphi_A(1) := 1$$
$$\varphi_A(n) := \#\{k \in \mathbb{Z}_n : \forall\ a \in A,\ (k + a, n) = 1\}$$

Theorem 2.2. *For any k-tuple A, φ_A is multiplicative.*

Proof. Throughout, let k_n be the class represented by k in $\mathbb{Z}_n$ for any n. Let A be a k-tuple, and let m and n be coprime natural numbers. For each natural number b, define the set $P(b)$ by

$$P(b) = \{k_b \in \mathbb{Z}_b : \forall\ a \in A,\ (k_b + a, b) = 1\}.$$

Then by definition $\#P(a) = \varphi_A(a)$.

Now let $\Gamma : P(mn) \longmapsto P(m) \times P(n)$ be given by

$$\Gamma(k_{mn}) = (k_m, k_n).$$

We will show each of the following:

1. Γ **is well defined.** Let $k_{mn} \in P(mn)$. Then $\forall\ a \in A,\ (k+a, mn) = 1$, so $\forall\ a \in A,\ (k + a, m) = 1$. Proceeding likewise for n, $(k_m, k_n) \in P(m) \times P(n)$, so Γ is well defined.

2. Γ **is injective.** Let $a_{mn}, b_{mn} \in P(mn)$ such that $\Gamma(a_{mn}) = \Gamma(b_{mn})$. Then $a_m = b_m$ and $a_n = b_n$. Because m and n are coprime, $a_{mn} = b_{mn}$. Thus, Γ is injective.

3. Γ **is surjective.** Let $(b_m, c_n) \in P(m) \times P(n)$. By the Chinese Remainder Theorem, there exists $d_{mn} \in \mathbb{Z}_{mn}$ such that $d_m = b_m$ and $d_n = c_n$. It then suffices to show $d_{mn} \in P(mn)$, for this gives $\Gamma(d_{mn}) = (b_m, c_n)$.

   Let $a \in A$. We have

   $$\begin{aligned} & b_m \in P(m) \\ \implies & (a + b_m, m) = 1 \\ \implies & (a + d_m, m) = 1 \\ \implies & (a + d_{mn}, m) = 1. \end{aligned}$$

   Proceeding likewise for c_n, we have $(a + d_{mn}, n) = 1$. Combining these gives $(a + d_{mn}, mn) = 1$, so $d_{mn} \in P(mn)$ and Γ is surjective.

Thus, Γ is a bijection, so

$$\begin{aligned} \varphi_A(mn) &= \#P(mn) \\ &= \#(P(m) \times P(n)) \\ &= \#P(m) \cdot \#P(n) \\ &= \varphi_A(m) \cdot \varphi_A(n). \end{aligned}$$

□

3 The Unit Function

The singular unit function $u(n)$ is defined by $u(n) = 1$ for all n. It is surprising that such a trivial function has a non-trivial extension to k-tuples.

Definition 3.1. *For any k-tuple A and prime p, define $u_A(p)$ by:*

$$u_A(p) = \#A_p$$

We extend u_A to all natural numbers by complete multiplicativity.

We have immediately from complete multiplicativity that $u_A(1) = 1$. The definition for primes tells us that u_A gives the number of residues the elements of A fill modulo p. The definition of u_A on primes is used in the first Hardy and Littlewood conjecture, discussed later, and so is not entirely new. It's usage in the following theorem to describe extensions of φ is also known, at least in some well studied cases.

Theorem 3.2. *For any k-tuple A, prime p, and natural number k we have:*

$$\varphi_A(p^k) = p^k - u_A(p)p^{k-1}.$$

Proof. By definition we have

$$\begin{aligned}\varphi_A(p^k) &= \#\{j \in \mathbb{Z}_{p^k} : \forall\ a \in A,\ (j+a, p^k) = 1\} \\ &= \#\{j \in \mathbb{Z}_{p^k} : \forall\ a \in A_{p^k},\ (j+a, p^k) = 1\}.\end{aligned}$$

If this condition is satisfied by some j, it will be satisfied by $j + pm$ for any m, and there are p^{k-1} multiples of p in $\mathbb{Z}_{p^k}$. In other words, as A_{p^k} is translated through $\mathbb{Z}_{p^k}$, we gain the same amount towards our total count as we pass through each period of size p, and there are p^{k-1} such periods. We then have

$$\begin{aligned}\varphi_A(p^k) &= p^{k-1}\#\{j \in \mathbb{Z}_p : \forall\ a \in A_p,\ (j+a, p) = 1\} \\ &= p^{k-1}(p - \#\{j \in \mathbb{Z}_p : \exists\ a \in A_p,\ (j+a, p) \neq 1\})\end{aligned}$$

Now, $(j + a, p) \neq 1$ exactly once per $a \in A_p$ as j ranges over $\mathbb{Z}_p$ (when $j + a \equiv 0 \bmod p$). Thus, the size of the set in the previous equation is $\#A_p$, which gives the required formula. $\square$

Theorem 3.3 (Totient Product Formula)**.**
For any k-tuple A and natural number n, we have:

$$\varphi_A(n) = n \prod_{p|n} \left(1 - \frac{u_A(p)}{p}\right).$$

Proof. Let A be a k-tuple and n be a natural number. Then n has a factorization, say

$$n = \prod_{p_i|n} p_i^{a_i}.$$

Then we have by theorems 2.2 and 3.2,

$$\begin{aligned}
\varphi_A(n) &= \varphi_A\left(\prod_{p_i|n} p_i^{a_i}\right) \\
&= \prod_{p_i|n} \varphi_A(p_i^{a_i}) \\
&= \prod_{p_i|n} \left(p_i^{a_i} - u_A(p_i) p_i^{a_i - 1}\right) \\
&= n \prod_{p_i|n} \left(1 - \frac{u_A(p_i)}{p_i}\right).
\end{aligned}$$

□

Theorem 3.4. *For any k-tuple A, the totient function has the following properties:*

(a) $\varphi_A(mn) = \varphi_A(m)\varphi_A(n)(d/\varphi_A(d))$, *where* $d = (m, n)$.

(b) $\varphi_A(mn) = \varphi_A(m)\varphi_A(n)$ *if* $(m, n) = 1$.

(c) $a|b$ *implies* $\varphi_A(a)|\varphi_A(b)$.

Proof. The proof of these follows nearly verbatim the singular case, which is shown in Theorem 2.5 of [1].

□

Definition 3.5. *Let v be an arithmetic function such that:*

1. *v is completely multiplicative.*
2. *v is eventually constant, say $v(p) = k$ for $p > N$.*

3. For all primes p, $v(p) \leq p$ and $v(p) \leq k$.

*Then we say v is **unit looking**. If there exists a k-tuple such that $u_A = v$, we say A generates v, and say v is good. Otherwise, v is bad.*

Unit looking functions, as far as can be checked by simple methods, appear to have been generated by some k-tuple. For $k = 2$, all unit looking functions are good. Proving any individual unit looking function is bad is a tedious affair involving many cases, and so is omitted. A general algorithm determining this property would be useful in later computation, but it seems unlikely such an algorithm is feasable in the general case. As a concrete example, the unit looking function v with $v(2) = 1$, $v(3) = v(103) = 2$, and $v(p) = 3$ for all other primes p is bad, and is the minimal example. Computation shows that bad unit looking functions become far more common as the primes at which they differ from their eventual value grow, and it seems likely that the proportion of good functions goes to zero in that limit. We may, however, easily find k-tuples whose unit matches a given unit looking function up to an arbitrary bound.

Theorem 3.6. *Let p_n be the nth prime number. Let v be a unit looking function. Then for any natural N, there exists a k-tuple A such that $u_A(n) = v(n)$ for $n \leq N$.*

Proof. By the complete multiplicativity of u_A and v, we need only show the equality on primes up to N.

Let k be the eventual value of v on primes. Let $A_1 = \{0\}$. If $k = 1$, we have $u_{A_1} = v$ and are done. Now for each i from 2 to k, let Q_i be the set of primes up to N for which $u_{A_i}(p) < v(p)$. By the Chinese Remainder Theorem there exists a which solves $a \equiv i \pmod{q}$ for all $q \in Q_i$, and $a \equiv 0$ modulo each other prime up to N. Let $A_i = A_{i-1} \cup a$.

Then for each prime p_n up to N, exactly the first $v(p_n)$ residues have been filled by elements of A_k. That is, $u_{A_k}(p_n) = v(p_n)$. $\square$

4 Dirichlet Convolution

For any arithmetic functions f and g we denote their Dirichlet product by

$$(f * g)(n) := \sum_{ab=n} f(a)g(b).$$

We denote the Dirichlet inverse of an arithmetic function f by f^{-1}. We define the arithmetic function N by $N(n) = n$.

Definition 4.1. *For any k-tuple A, define μ_A to be the unique Dirichlet inverse of u_A.*

By definition u_A is completely multiplicative, so by Theorem 2.17 in [1], we have its Dirichlet inverse to be

$$\mu_A(n) = u_A^{-1}(n) = \mu(n)u_A(n).$$

We have also from Theorem 2.18 of [1] that for any multiplicative function f and natural number n, that

$$\sum_{d|n} \mu(d)f(d) = \prod_{p|n}(1 - f(p)), \tag{1}$$

which is used to prove the following.

Theorem 4.2. *For any k-tuple A, we have*

$$u_A * \varphi_A = N.$$

Proof. Applying equation (1) with $f(n) = u_A(n)/n$ to the totient product formula, we have

$$\begin{aligned}
\varphi_A(n) &= n \prod_{p|n} \left(1 - \frac{u_A(p)}{p}\right) \\
&= n \sum_{d|n} \mu(d) \frac{u_A(d)}{d} \\
&= \sum_{d|n} \mu_A(d) \frac{n}{d} \\
&= (\mu_A * N)(n).
\end{aligned}$$

Taking the Dirichlet product of both sides by u_A, we obtain the theorem. □

This result, while simple, is the reason for our later analysis. Intuitively, φ_A should encode information about relative primality in the k-tuple sense. The function N is neutral in this regard, so we should expect $u_A = N * \varphi_A^{-1}$ to retain some aspect of that information while playing nicely with Dirichlet series.

5 Alternatives to φ_A

When we defined φ_A, we counted occurences of complete coprimality. We could also ask for the occurences of any amount of coprimality, a specific amount, or a minimum amount, among other requirements. However, these examples do not produce multiplicative φ analogs.

There is also a subtlety in defining them that doesn't exist in our φ_A. We defined φ_A in terms of translates of A, but we could have equivalently counted the translates of A_n. These definitions are inequivalent for the examples listed above, for we will have $\#A_n < \#A$ when elements of A coincide modulo n. That is, the issue with those examples is that they rely on the number of $a \in k + A$ or $a \in k + A_n$ to satisfy a property, which may differ. We may avoid this issue be only considering functions which count those k such that each of the elements of $k + A$ possesses some property, rather considering a property on $k + A$ itself. But to consider a function to be an analog of φ, the property considered should be "Are these elements coprime?". There is only one function of this type which considers this property: φ_A.

We may relax our constraints, though, on what property constitutes a totient to depend on the value of $(k + a, n)$, rather than if it is 1 or not. This gives not only additional k-tuple totients, but also additional singular totients. In general, we may use the property that $(k + a, n)$ is limited to a specific subset of the primes and obtain a multiplicative totient.

Definition 5.1. *Given a set of primes B, let $a \prec B$ mean that a is composed only of elements of B.*

In particular, if this set is exactly the primes less than some bound b, we are requiring $(k + a, n)$ to be b-smooth:

Definition 5.2. *If the prime factors of an integer k are bounded by b, we say k is b-**smooth**.*

Definition 5.3. *For a k-tuple A and a set of primes B, define $\varphi_{A,B}$ by*

$$\begin{aligned}
&\varphi_{A,B}(1) := 1, \\
&\varphi_{A,B}(n) := \#\{k \in \mathbb{Z}_n : \forall\ a \in A,\ (k + a, n) \prec B\}.
\end{aligned}$$

Theorem 5.4. *For any k-tuple A and set of primes B, $\varphi_{A,B}$ is multiplicative.*

Proof. The proof follows exactly that of φ_A and so is omitted. □

What, then, are the corresponding unit functions $u_{A,B}$? They are the same as our earlier u_A, except they are zero at every number containing a factor in B. They are completely multiplicative, can be used to compute $\varphi_{A,B}$ in the same way, and yield $u_{A,B} * \varphi_{A,B} = N$. The proofs of these facts follow exactly those for u_A. We return now to the realm of φ_A, keeping in mind that many results may be extended trivially in this way to $\varphi_{A,B}$.

The earlier nonmultiplicative examples are not without value, and not beyond hope. In essentially all cases, we can express the desired function as a linear combination of φ_A. Consider first for some n and k-tuple A, counting the $k \in \mathbb{Z}_n$ such that exactly q elements of $k + A$ are coprime to n. We call the value q the quota, and refer to this as the unwrapped, exact quota case.

Definition 5.5 (Quota k-tuple totients)**.** *For any k-tuple A and quota $q \in \mathbb{Z}$ with $0 \leq q \leq \#A$, define $\Upsilon_A^{q=}$ by*

$$\begin{aligned}
&\Upsilon_A^{q=}(1) := 1 \\
&\Upsilon_A^{q=}(n) := \#\{k \in \mathbb{Z}_n : q = \#\{a \in A : (k + a, n) = 1\}\}.
\end{aligned}$$

Replacing A with A_n in this definition yields a wrapped case, and requiring at least q values to be coprime to n yields a minimum quota case. We denote the wrapped cases with ϱ rather than Υ, and minimum quotas use $q \leq$ in the superscript. Clearly, $\Upsilon_A^{\#A=} = \varphi_A$, though this is also proven by the following theorem.

Theorem 5.6. *For any k-tuple A and quota q,*

$$\Upsilon_A^{q=} = \sum_{k=q}^{\#A} (-1)^{k-q} \binom{q}{k} \sum_{\substack{B \subset A \\ \#B = k}} \varphi_B$$

Proof. TODO □

Theorem 5.7. *For any k-tuple A, and quota q,*

$$\Upsilon_A^{q\leq} = \sum_{k=q}^{\#A} (-1)^{k-q} \binom{k-1}{k-q} \sum_{\substack{B \subset A \\ \#B = k}} \varphi_B.$$

Proof. TODO □

The formulae for the corresponding ϱ functions are identical: simply replace A with A_n throughout. This creates a dependance on n in the final sum which appears to explode the computational complexity, but for sufficiently large n, $A = A_n$, so the order of computational complexity is the same. This also shows that the wrapped and unwrapped totients are eventually equivalent, as should be expected. We will make no further use of the wrapped totients, and the proofs are mostly the same as the unwrapped cases, so they are omitted.

Using the discrete modular convolution, we can obtain the essential data required to compute wrapped quota totients for all quotas at once.

Definition 5.8. *Given $f, g : \mathbb{Z}_n \to \mathbb{C}$, define the discrete modular convolution $\odot_n$ by*

$$(f \odot_n g)(k) = \sum_{\substack{a+b \equiv k \\ \bmod n}} f(a)g(b),$$

where $a, b \in \mathbb{Z}_n$.

The symbol $\odot$ is non-standard. Normally $*$ would be used, but it is already in use for Dirichlet convolution.

Theorem 5.9. *Let $A \in \mathbb{K}$, let $n \in \mathbb{N}$, and let χ_0 be the principle Dirichlet character mod n. Let 1_{A_n} be the indicator function of A_n. Then*

$$(1_{A_n} \odot_n \chi_0)(k) = \#\{a \in A_n : (k + a, n) = 1\}.$$

Proof. TODO □

Computing $1_{A_n} \odot_n \chi_0$ for all k then gives a convenient data format for querying quota totient information at n. This can be made to work for unwrapped quota totients by considering a period of size n of the non-modular discrete convolution of χ_0 and 1_A.

6 Admissibility

Definition 6.1. *We say a k-tuple A is **admissible** if, for each prime p, we have*

$$u_A(p) < p.$$

Then a k-tuple is admissible if it does not fill all of the residue classes modulo any prime. This condition gives a simple bound on u_A.

Proposition 6.2. *For any admissible k-tuple A, we have*

$$u_A(n) \leq \varphi(n).$$

Proof. Let $A \in \mathbb{K}$ be admissible, and let n be a natural number with factorization

$$n = \prod_{p_i|n} p_i^{a_i}.$$

Then

$$\begin{aligned} u_A(n) &= u_A\left(\prod_{p_i|n} p_i^{a_i}\right) \\ &= \prod_{p_i|n} u_A(p_i)^{a_i} \\ &\leq \prod_{p_i|n} (p_i - 1)^{a_i} \\ &\leq \prod_{p_i|n} (p_i - 1) \cdot p_i^{a_i - i} \\ &= \varphi(n). \end{aligned}$$

□

We are now equipped to state the prime k-tuple conjecture.

Conjecture 6.3 (Prime K-Tuple Conjecture)**.** *For any finite, admissible k-tuple A, there exists an infinite k-tuple B such that $A + B \subset \mathbb{P}$.*

Admissibility is required, for if a k-tuple A was inadmissible by prime p, for $b \in B$ large enough that each element of $b + A$ is greater than p, we must have some element of A filling the 0 residue. That element is then divisible by p, $b + A \not\subset \mathbb{P}$, and it cannot be for any b beyond that bound. Thus, there can be only finitely many integers at which $b + A \subset \mathbb{P}$ and no valid B exists.

The finiteness of A is also required. For example, let A be the set of all primorials. This fills at most i residues of each prime p_i and so is admissible. We cannot have negatives in B, for $2 \in A$ is the minimum prime. We cannot have 0 or 1 in B, because $A + B$ would contain 6 and 4 respectively. If B contains $b > 1$, b is divisible by some prime p, and the primorials will eventually be divisible by p. Then for some element $a \in A$ divisible by p, $p|a + b$, so $A + B \not\subset \mathbb{P}$. Thus, there are no integers b such that $b + A \subset \mathbb{P}$. Not only can B not be infinite in this case, it cannot contain any integer.

If the conjecture holds, there exist infinite k-tuples A with arbitrary large finite B such that $A+B \subset \mathbb{P}$. Indeed, given a finite admissible k-tuple B, we apply the conjecture and find such an A. The question of adding two infinite k-tuples and obtaining a prime k-tuple is handled in greater generality in [2], the results of which we state in less generality as the following theorem. It is often said that the Prime K-Tuple Conjecture feels intuitively true, so it is somewhat alarming that it holds as a consequence the following result, which seems intuitively impossible.

Theorem 6.4. *Assume the Prime K-Tuple Conjecture. Then there exist infinite k-tuples A and B such that $A + B \subset \mathbb{P}$.*

7 The First Hardy-Littlewood Conjecture

Definition 7.1. *For any admissible k-tuple A, define*

$$\pi_A(x) = \#\{n \leq x : n + A \subset \mathbb{P}\}.$$

Definition 7.2. *For $n > 0$, define*

$$\mathrm{Li}_n(x) = \int_2^\infty \frac{\mathrm{dt}}{\log^n(t)}.$$

We call this the nth (offset) logarithmic integral.

Conjecture 7.3 (First Hardy-Littlewood conjecture)**.** *Let A be an admissible k-tuple. Then*

$$\pi_A(x) \sim C_A \mathrm{Li}_{\#A}(x),$$

where

$$C_A = \prod_p \frac{1 - u_A(p)p^{-1}}{(1 - p^{-1})^{\#A}}.$$

This, of course, implies the prime k-tuple conjecture. It also implies that the exact structure of a k-tuple does not determine its asymptotic behavior, but rather that behavior is entirely decided by u_A. For example, $\{0, 2\}$ and $\{0, 4\}$ produce the same u_A. This yields an equivalence relation.

Definition 7.4. *Given k-tuples A and B, we say they are* ***residually equivalent*** *if $u_A = u_B$.*

The following asymptotic expansion is well known:

$$\mathrm{Li}_1(x) \sim \frac{x}{\log(x)} \sum_{k \geq 0} \frac{k!}{\log^k(x)}. \tag{2}$$

The simpler estimate $\frac{x}{\log x}$ given in the prime number theorem is then the truncation of this expansion to its first term. Then we should expect to analagously use the truncation of the asymptotic expansion of Li_n in attempting to gain traction in the first Hardy-Littlewood conjecture.

Proposition 7.5. *TODO: this result needs additional care due to the meaning of asymptotic expansion, though it is correct. We could also just state it as true, I don't think proving this is all that important. We have the asymptotic expansion of* Li_n *to be*

$$\mathrm{Li}_n(x) \sim \frac{x}{(n-1)! \log x} \sum_{k \geq n-1} \frac{k!}{\log^k(x)}.$$

Proof. First, apply integration by parts to Li_n with $dv = dx/(x \log^n(x))$ and $u = x$ to obtain

$$\mathrm{Li}_n(x) = \frac{-x}{(n-1)\log^{n-1} x} + \frac{1}{n-1}\mathrm{Li}_{n-1}(x) \tag{3}$$

for $n > 1$. We have the result for $n = 1$: it is exactly equation (2). Assume then for induction the result holds for some n. Then applying equation (3) we have

$$\begin{aligned}
\mathrm{Li}_{n+1}(x) &= \frac{-x}{n \log^n x} + \frac{1}{n}\mathrm{Li}_n(x) \\
&\sim \frac{-x}{n \log^n x} + \frac{x}{n! \log x} \sum_{k \geq n-1} \frac{k!}{\log^k x} \\
&= \frac{x}{n! \log x} \left(\frac{-(n-1)!}{\log^{n-1} x} + \sum_{k \geq n-1} \frac{k!}{\log^k x} \right) \\
&= \frac{x}{n! \log x} \sum_{k \geq n} \frac{k!}{\log^k x}.
\end{aligned}$$

□

All this is to say: if the first Hardy-Littlewood conjecture is correct, we should attempt to show, following in the steps of the prime number theorem, that

$$\pi_A(x) \sim \frac{C_A x}{\log^{\#A} x}.$$

8 The Dirichlet Series of u_A

We denote the Derichlet series of an arithmetic function f by

$$\mathcal{D}(f)(s) := \sum_{n\geq 1} \frac{f(n)}{n^s}.$$

For each admissible constellation A, Conjecture 7.3 gives an associated constant C_A. We may factor $\mathcal{D}(u_A)$ as

$$\mathcal{D}(u_A)(s) = \zeta^{\#A}(s) \prod_p \frac{(1-p^{-s})^{\#A}}{1-u_A(p)p^{-s}}. \tag{4}$$

We refer to the final product as the modulator, denoted $N_A(s)$. The following result essentially states that the product which yields C_A converges, which is well known, but is given for completeness.

Proposition 8.1. *For any admissible k-tuple A, there exists $\epsilon > 0$ such that $N_A(s)$ converges for $\Re(s) > 1-\epsilon$.*

Proof. Let A be an admissible k-tuple. Then $u_A(p) < p$, so the real parts of the solutions of $1 - u_A(p)p^{-s} = 0$ are all left of 1, and go to 0 as $p \to \infty$. Then there exists $\epsilon > 0$ such that all solutions of $1 - u_A(p)p^{-s} = 0$ lie to the left of $1-\epsilon$. Fix $s = \sigma + it$ with $\sigma > 1/2$. Let

$$N_{A,r}(s) = \prod_{p>r} \frac{(1-p^{-s})^{\#A}}{1-u_A(p)p^{-s}}.$$

We will show the convergence of $N_{A,r}$ at s for a fixed r, from which the result follows, as the finitely many removed factors all converge for $\Re(s) > 1-\epsilon$. We fix r to be large enough that both $\#Ar^{-\sigma} < 1/2$, and $u_A(p) = \#A$ for $p > r$. We then have

$$\begin{aligned}\log N_{A,r}(s) &= \log \prod_{p>r} \frac{(1-p^{-s})^{\#A}}{1-\#Ap^{-s}} \\ &= \sum_{p>r} \left(\#A \log(1-p^{-s}) - \log(1-\#Ap^{-s})\right).\end{aligned}$$

Expanding the logs into power series gives

$$\begin{aligned}\log N_{A,r}(s) &= \sum_{p>r} \left(-\#A \sum_{k\geq 1} \frac{p^{-sk}}{k} + \sum_{k\geq 1} \frac{(\#Ap^{-s})^k}{k} \right) \\ &= \sum_{p>r} \sum_{k\geq 2} \left(\frac{(\#A)^k - \#A}{k} p^{-sk} \right).\end{aligned}$$

Note the cancellation of the $k = 1$ term. Now,

$$|\log N_{A,r}(s)| \leq \sum_{p>r} \sum_{k\geq 2} \left(\#Ap^{-\sigma}\right)^k.$$

For each $p > r$, by our bound on r, the k sum is a convergent geometric series. We have also for $p > r$ that $1 - \#Ap^{-\sigma} > 1/2$, so

$$\begin{aligned} |\log N_{A,r}(s)| &= \sum_{p>r} \frac{(\#Ap^{-\sigma})^2}{1 - \#Ap^{-\sigma}} \\ &\leq 2(\#A)^2 \sum_{p>r} p^{-2\sigma} \\ &\leq 2(\#A)^2 \sum_{n>r} n^{-2\sigma}. \end{aligned}$$

We have $\sigma > 1/2$, so this is a convergent p-series. Thus, $\log N_{A,r}$ converges absolutely, and exponentiating gives the result. □

We do not need the full domain of convergence of the prior result; our goal was to justify evaluating $N_A(1)$ and obtaining the following.

Corollary 8.2. *For any admissible k-tuple A, we have*

$$\frac{1}{C_A} = N_A(1),$$

and

$$Res(\zeta^{\#A}, 1) = C_A Res(\mathcal{D}(u_A), 1).$$

Proof. The first equality is immediate from the definitions of C_A and N_A. Taking residues in (4) then yields the second equality. □

The Laurent expansion of ζ is known, so the residues above are computable. We may factor N_A further into $\mathcal{D}((\#A)^{\Omega})$ and a finite product, but for k-tuples with a prime size, this creates a pole and zero at $s = 1$ in those factors respectively. We now compute $N_A(1)$ for some admissible, non-residually-equivalent k-tuples, where the p column gives those primes over which $u_A(p) \neq \#A$. Computations were made using the product over the first 10000 primes before rounding, and are made only for the sake of discussion: they should not be used outside this context. We also include the corresponding C_As.

A	p	$N_A(1)$	C_A
$\{0,2\}$	2	0.75738949	1.32032464
$\{0,10\}$	2,5	0.56804211	1.76043285
$\{0,6\}$	2,3	0.37869474	2.64064928
$\{0,2,6,8\}$	2,3	0.24089421	4.15119988
$\{0,8,12,30\}$	2,3,5,11	0.10539121	9.48845686
$\{0,6,12,18\}$	2,3	0.12044710	8.30239975
$\{0,2,6,8,30\}$	2,3,5,7,11	0.05639911	17.73077650
$\{0,18,30,60\}$	2,3,5,7	0.03011178	33.20959900
$\{0,60,120,180\}$	2,3,5	0.03011178	33.20959900

We see that the final two entries appear to be exactly equal. It is not a rounding error: they are identical. This, along with their corresponding k-tuples having the same size, means that the Hardy-Littlewood Conjecture predicts these k-tuples to have identical limiting frequency, despite their residual inequivalence.

When k-tuples A and B have the same size, the ratio $N_A(s)/N_B(s)$ is a finite product, so $N_A(1)/N_B(1)$ is rational. It is convenient to handle these constants in a rational form, so we would like to define, for each k, a constant representing the infinite portion of the product. The naive candidate is $\prod_p \frac{(1-p^{-1})^{\#A}}{1-p^{-s}}$, but this diverges. We can use $N_A(1)$ for any particular A, but opt to instead choose the maximum of such constants.

Definition 8.3. *For each $k \in \mathbb{N}$, define*

$$\mathfrak{U}_k = \max\{N_A(1) : \#A = k\}.$$

Proposition 8.4. *For any $k \in \mathbb{N}$, we have*

$$\mathfrak{U}_k = N_A(1)$$

for A containing ???

Proof. I think it is OEIS: A135311. Verify via simulation. □

Some of these minimal constants are given below.

k	$\mathfrak{U}_k$
1	0
2	0
3	0
4	0
5	0
6	0
7	0
8	0
9	0

We may express the limiting behavior of the summatory function of u_A via $N_A(1)$.

Theorem 8.5. *For any admissible k-tuple A, we have*

$$\frac{1}{x \log^{\#A-1} x} \sum_{n \leq x} u_A(n) \sim \frac{N_A(1)}{(\#A-1)!}.$$

Proof. This is exactly the result of Theorem 5.2 in [3]. □

9 TODO

1. We have $u_A * \phi_A = N = \phi_A^0$. That is, u_A jumps between levels of phi. What functions jump between other levels?

2. Can the quota totient formulae be inverted to express higher quotas in terms of lower? Are there corresponding formulae relating unit functions for quota totients?

3. Double check Granville actually implies the thing. Oh, just noticed he is the pretentious theory guy.

4. We should check if any results require finiteness on k-tuple. Also check each theorem for admissibility requirements.

References

[1] Tom M. Apostol. *Introduction to Analytic Number Theory*. Springer, 1976. ISBN: 9788184895216.

[2] Andrew Granville. *A note on the sums of primes*. Tech. rep. University of Toronto, 1990.

[3] Gerald Tenenbaum. *Introduction to Analytic and Probabilistic Number Theory*. American Mathematical Society, 2008. ISBN: 9781470478216.